

# How to use these token cutouts



These pages contain the text *The Tale of Peter Rabbit* by Beatrix Potter. Each **cutout** shows a **next word** paired with the **previous word** that came immediately before it in the original text.

| Model vitals             |                                   |
|--------------------------|-----------------------------------|
| <b>1123</b><br>tokens    | <b>2.25</b><br>bits/token entropy |
| <b>4.8</b><br>perplexity | <b>2.32</b><br>branching factor   |

## Anatomy of a cutout



Every distinct word has its own colour. **Previous** words appear inside a coloured box (the word's own colour as the background, with white text); the free-standing **next word** appears in plain coloured text. The same word always wears the same colour, whether you see it inside a box or free-standing. Two unrelated words can occasionally share a colour, so always verify the word itself matches—not just the colour.

## Setup

**Cut out the tokens** along the dotted lines and **spread them out** face-up on the table, with no overlap if possible. This is the “training” step: every (previous, next) combination from the original text is now a physical cutout sitting on your table.

## How to play: the matching game

To generate text, repeatedly find a cutout whose **previous word** matches the last word you've written, then **write its next word** onto your page. The word you just wrote becomes part of what you'll match against next—that's the chain.

## Worked example

**Step 1.** Pick any cutout to begin—say this one—and copy its previous word *and* its next word onto your page.

**Step 2.** Look at the last word you’ve written (*upon*) and find a cutout whose previous word matches it. Write its next word.

**Step 3.** Repeat: find a cutout whose previous word matches the last word you’ve written, and add its next word.

Keep going—write as much or as little as you like.

*the cutout*

Once upon

upon a

a time

*your page*

Once upon

Once upon a

Once upon a **time**

## Tips

**Use colour as a fast filter.** Scan by the rightmost previous-word box’s colour first, then verify the actual word matches.

**Pick from many candidates by eye.** The more cutouts share the same previous word, the more often your eye lands on one—that’s **weighted sampling** for free, and it’s why some words follow others more often in your text.

**Put the cutout back** after using it—removing it would change the model’s distribution next time round.

Once **Once** upon **upon** a **a** time **time** there  
**there** were **were** four **four** little **little** Rabbits  
**Rabbits** , **,** and **and** their **their** names  
**names** were **were** Flopsy **Flopsy** , **,** Mopsy  
**Mopsy** , **,** Cotton **Cotton** tail **tail** , **,** and  
**and** Peter **Peter** . **.** They **They** lived **lived** with  
**with** their **their** mother **mother** in **in** a **a** sand  
**sand** bank **bank** , **,** underneath **underneath** the  
**the** root **root** of **of** a **a** very **very** big **big** fir  
**fir** tree **tree** . **.** now **now** my **my** dears **dears** ,

, said said old old Mrs Mrs . . rabbit

rabbit one one morning morning , , you

you may may go go into into the the fields

fields or or down down the the lane lane ,

, but but don't don't go go into into Mr Mr .

. McGregor's McGregor's garden garden : : your

your Father Father had had an an accident

accident there there ; ; he he was was put

put in in a a pie pie by by Mrs Mrs .

. McGregor McGregor . . now now run

**run** along **along** , and **and** don't **don't** get

**get** into **into** mischief **mischief** . **.** I **I** am

**am** going **going** out **out** . **.** then **then** old

**old** Mrs **Mrs** . **.** rabbit **rabbit** took **took** a

**a** basket **basket** and **and** her **her** umbrella

umbrella , and **and** went **went** through

**through** the **the** wood **wood** to **to** the

**the** baker's **baker's** . **.** she **she** bought **bought** a

**a** loaf **loaf** of **of** brown **brown** bread **bread** and

**and** five **five** currant **currant** buns **buns** .

. Flopsy Flopsy , Mopsy Mopsy , and  
and Cottontail Cottontail , who who were  
were good good little little bunnies bunnies ,  
, went went down down the the lane lane to  
to gather gather blackberries blackberries :  
: but but Peter Peter , who who was  
was very very naughty naughty , ran  
ran straight straight away away to to Mr Mr .  
. McGregor's McGregor's garden garden , and  
and squeezed squeezed under under the

the gate gate ! ! first first he he ate ate some

some lettuces lettuces and and some

some French French beans beans ; ; and

and then then he he ate ate some

some radishes radishes ; ; and and then then ,

, feeling feeling rather rather sick sick , , he

he went went to to look look for for some

some parsley parsley . . but but round

round the the end end of of a a cucumber

cucumber frame frame , , whom whom should

**should** he **he** meet **meet** but **but** Mr **Mr** .

. McGregor **McGregor** ! **!** Mr **Mr** . . McGregor

**McGregor** was **was** on **on** his **his** hands

**hands** and **and** knees **knees** planting

**planting** out **out** young **young** cabbages

**cabbages** , **,** but **but** he **he** jumped **jumped** up

**up** and **and** ran **ran** after **after** Peter **Peter** ,

**,** waving **waving** a **a** rake **rake** and **and** calling

**calling** out **out** , **,** Stop **Stop** thief **thief** !

**!** Peter **Peter** was **was** most **most** dreadfully



**dreadfully** frightened **frightened** ; he

he **rushed** **rushed** all **all** over **over** the

**the** garden **garden** , **for** **for** he **he** had

had **forgotten** **forgotten** the **the** way **way** back

back **to** **to** the **the** gate **gate** . he **he** lost

lost one **one** of **of** his **his** shoes **shoes** among

among the **the** cabbages **cabbages** , and

and the **the** other **other** shoe **shoe** amongst

**amongst** the **the** potatoes **potatoes** . after

after losing **losing** them **them** , he **he** ran

ran on on four four legs legs and and went

went faster faster , , so so that that I I think

think he he might might have have got

got away away altogether altogether if if he

he had had not not unfortunately

unfortunately run run into into a a gooseberry

gooseberry net net , , and and got got caught

caught by by the the large large buttons

buttons on on his his jacket jacket . . it

it was was a a blue blue jacket jacket with

with brass brass buttons buttons , , quite

quite new new . . Peter Peter gave

gave himself himself up up for for lost lost ,

, and and shed shed big big tears tears ; ; but

but his his sobs sobs were were overheard

overheard by by some some friendly

friendly sparrows sparrows , , who who flew

flew to to him him in in great

great excitement excitement , , and

and implored implored him him to to exert

exert himself himself . Mr Mr . McGregor

McGregor came came up up with with a

a sieve sieve , , which which he he intended

intended to to pop pop upon upon the the top

top of of Peter Peter ; ; but but Peter

Peter wriggled wriggled out out just just in

in time time , , leaving leaving his his jacket

jacket behind behind him him . . and

and rushed rushed into into the the tool

tool shed shed , , and and jumped

**jumped** into **into** a **a** can **can** . **it** **it** would

**would** have **have** been **been** a **a** beautiful

**beautiful** thing **thing** to **to** hide **hide** in **in** ,

**,** **if** **if** it **it** had **had** not **not** had **had** so

**so** much **much** water **water** in **in** it **it** . **Mr**

**Mr** . **McGregor** **McGregor** was **was** quite

**quite** sure **sure** that **that** Peter **Peter** was

**was** somewhere **somewhere** in **in** the **the** tool

**tool** shed **shed** , **,** perhaps **perhaps** hidden

**hidden** underneath **underneath** a **a** flower

flower pot pot . he he began began to  
to turn turn them them over over carefully  
carefully , , looking looking under under each  
each . . presently presently Peter Peter sneezed  
sneezed Kertyschoo Kertyschoo ! ! Mr Mr .  
. McGregor McGregor was was after after him  
him in in no no time time . . and and tried  
tried to to put put his his foot foot upon  
upon Peter Peter , , who who jumped  
jumped out out of of a a window window ,

, upsetting **upsetting** three **three** plants **plants** .

. the **the** window **window** was **was** too

**too** small **small** for **for** Mr **Mr** . **McGregor**

**McGregor** , , and **and** he **he** was **was** tired

**tired** of **of** running **running** after **after** Peter

**Peter** . **he** **he** went **went** back **back** to **to** his

**his** work **work** . **Peter** **Peter** sat **sat** down

**down** to **to** rest **rest** ; **;** he **he** was **was** out

**out** of **of** breath **breath** and **and** trembling

**trembling** with **with** fright **fright** , , and



and he he had had not not the the least

least idea idea which which way way to to go

go . . Also Also he he was was very

very damp damp with with sitting sitting in

in that that can can . . after after a a time

time he he began began to to wander

wander about about , , going going lippity

lippity lippity lippity not not very very fast

fast , , and and looking looking all all round

round . . he he found found a a door door in



in a a wall wall ; ; but but it it was

was locked locked , , and and there there was

was no no room room for for a a fat fat little

little rabbit rabbit to to squeeze

squeeze underneath underneath . . an an old

old mouse mouse was was running running in

in and and out out over over the the stone

stone doorstep doorstep , , carrying

carrying peas peas and and beans beans to

to her her family family in in the the wood

wood . Peter Peter asked asked her her the  
the way way to to the the gate gate , , but  
but she she had had such such a a large  
large pea pea in in her her mouth mouth that  
that she she could could not not answer  
answer . she she only only shook shook her  
her head head at at him him . Peter  
Peter began began to to cry cry . then  
then he he tried tried to to find find his  
his way way straight straight across across the

**the** garden **garden** , **,** but **but** he **he** became

**became** more **more** and **and** more

**more** puzzled **puzzled** . **.** presently **presently** ,

**,** he **he** came **came** to **to** a **a** pond **pond** where

**where** Mr **Mr** . **.** McGregor **McGregor** filled

**filled** his **his** water **water** cans **cans** . **.** a

**a** white **white** cat **cat** was **was** staring

**staring** at **at** some **some** gold **gold** fish **fish** ,

**,** she **she** sat **sat** very **very** , **,** very **very** still

**still** , **,** but **but** now **now** and **and** then

then the the tip tip of of her her tail

tail twitched twitched as as if if it it were

were alive alive . . Peter Peter thought

thought it it best best to to go go away

away without without speaking speaking to

to her her ; ; he he had had heard

heard about about cats cats from from his

his cousin cousin , , little little Benjamin

Benjamin Bunny Bunny . . he he went

went back back towards towards the the tool

tool shed shed , but but suddenly suddenly ,  
, quite quite close close to to him him , he  
he heard heard the the noise noise of of a  
a hoe hoe scr scr r ritch ritch , scratch  
scratch , scratch scratch , scritch scritch .  
. Peter Peter scuttered scuttered underneath  
underneath the the bushes bushes . but  
but presently presently , as as nothing  
nothing happened happened , he he came  
came out out , and and climbed

climbed upon upon a a wheelbarrow

wheelbarrow and and peeped peeped over

over . the the first first thing thing he

he saw saw was was Mr Mr . . McGregor

McGregor hoeing hoeing onions onions . . his

his back back was was turned turned towards

towards Peter Peter , , and and beyond

beyond him him was was the the gate gate !

! Peter Peter got got down down very

very quietly quietly off off the the wheelbarrow

**wheelbarrow** ; ; and **and** started

**started** running **running** as **as** fast **fast** as

**as** he **he** could **could** go **go** , , along **along** a

**a** straight **straight** walk **walk** behind

**behind** some **some** black **black** currant

**currant** bushes **bushes** . . Mr **Mr** . . **McGregor**

**McGregor** caught **caught** sight **sight** of **of** him

**him** at **at** the **the** corner **corner** , , but

but Peter **Peter** did **did** not **not** care **care** .

. he **he** slipped **slipped** underneath

underneath the the gate gate , , and and was

was safe safe at at last last in in the the wood

wood outside outside the the garden garden .

. Mr Mr . . McGregor McGregor hung hung up

up the the little little jacket jacket and

and the the shoes shoes for for a a scare

scare crow crow to to frighten frighten the

the blackbirds blackbirds . . Peter Peter never

never stopped stopped running running or

or looked looked behind behind him him till



till he he got got home home to to the  
the big big fir fir tree tree . he he was  
was so so tired tired that that he he flopped  
flopped down down upon upon the the nice  
nice soft soft sand sand on on the the floor  
floor of of the the rabbit rabbit hole hole and  
and shut shut his his eyes eyes . his  
his mother mother was was busy busy cooking  
cooking ; ; she she wondered wondered what  
what he he had had done done with with his

his clothes clothes . it it was was the  
the second second little little jacket jacket and  
and pair pair of of shoes shoes that that Peter  
Peter had had lost lost in in a a fortnight  
fortnight ! ! I I am am sorry sorry to to say  
say that that Peter Peter was was not not very  
very well well during during the the evening  
evening . his his mother mother put put him  
him to to bed bed , , and and made  
made some some camomile camomile tea tea ;

; and and she she gave gave a a dose dose of  
of it it to to Peter Peter ! ! one one table  
table spoonful spoonful to to be be taken  
taken at at bed bed time time . . but  
but Flopsy Flopsy , , Mopsy Mopsy , , and  
and Cotton Cotton tail tail had had bread  
bread and and milk milk and and blackberries  
blackberries for for supper supper .